

MNIST Handwritten Digits Classification — Deep Neural Networks

Author

Oklaliotis Pantelis-Panagiotis

- Hyperparameter Tuning

Variation	Layers	Neurons/Layer	Activation	Learning Rate	Epochs	Accuracy	Loss	Validation Accuracy	Validation Loss
A	2	512, 256	relu	0.001	10	0.9956	0.0195	0.9561	0.2127
B	3	512, 256, 128	relu	0.001	15	0.9992	0.0080	0.9600	0.1730
C	2	512, 256	relu	0.01	10	0.0989	NaN	0.0980	NaN
D	2	512, 256	sigmoid	0.001	10	0.9010	0.4226	0.9034	0.4010
E	2	128, 64	relu	0.001	10	0.9563	0.1470	0.9471	0.2030
F	3	256, 128, 64	relu	0.001	20	0.9950	0.0237	0.9573	0.2018

Many different parameters were tested in order to select the best ones for our network. The optimal approach based on the statistics we have is given by variation **B**:

- **Layers:** 3 (increases the ability to learn more complex patterns)
- **Neurons:** 512, 256, 128 (gradual reduction helps with regularization)
- **Activation:** **relu** (better performance than **tanh** and **sigmoid**)
- **Learning Rate:** 0.001 (stable learning without oscillations)
- **Epochs:** 15 (allows loss convergence)

Improvements

To improve the performance of the neural network beyond *Hyperparameter Tuning*, some improvements were applied to the network architecture and data:

1. **Input normalization to the range [0, 1]:** MNIST data has pixel values in the range [0, 255]. By normalizing them to be in [0, 1], we help the network learn faster and more stably.
2. **Add momentum=0.9 to SGD to accelerate the learning process.**
3. **Increase epochs to 30 to increase loss convergence.**
4. **Add Dropout layers (0.2) to reduce overfitting**
 - by randomly deactivating neurons during training.

Comprehension Questions

- a) MNIST data is considered good for training a model, especially for educational purposes and simple image classification applications. It contains 10 classes (0–9) with an approximately **equal number of samples per class**, and the images are small (28×28 px, grayscale), ideal for rapid experimentation. However, it also has certain limitations. The

dataset is quite simplistic, as the images are centered, clean, and do not reflect real-world "noise". In addition, they do not cover complex variability, such as distortions and rotations.

- b) No, **not all pixels are equally important** for predicting the class of a digit in MNIST. Certain pixels contain **useful information**, while others are **almost always "empty" or zero** and do not substantially contribute to the model's learning.
 - c) The use of **Deep Neural Networks** is a good idea **when the problem has high complexity and requires understanding complex or abstract patterns**. Some cases where DNNs excel are image processing, speech recognition, natural language processing, reinforcement learning, etc.
 - d) **Deep Learning** can be used **in all three main branches of Machine Learning**:
 - **Supervised learning**: Predicting whether an image contains a cat or a dog. We can use CNNs for image classification and Deep regression models for predicting continuous values.
 - **Unsupervised learning**: Grouping handwritten digits with clustering. We can use autoencoders for feature extraction and GANs for data generation.
 - **Reinforcement learning**: An agent playing chess and improving through experience. We can use DQNs for decision optimization and CNNs for input from images.
-

- **GitHub Repository**: [Link](#)
- **Google Colab Notebook**: [Link](#)